

master

Go to file

Add file

Code

About

	davidwillingham	minor update	...
	Images	new image for	
	LICENSE	Initial Commi	
	MATLABDeepLearningMod...	minor update	
	README.md	minor update	
	SECURITY.md	Initial Commi	
	viewDeepLearningModelH...	Script for laur	

Local

Codespaces New

Clone

HTTPSSSHGitHub CLI

https://github.com/matlab-deep-learning/MATLA

Use Git or checkout with SVN using the web URL.

Open with GitHub Desktop

Download ZIP

Discover pretrained models for deep learning in MATLAB

[www.mathworks.com/solutions/de...](#)

[#deep-learning](#) [#model-zoo](#) [#matlab](#)

[#pretrained-models](#) [#matlab-deep-learning](#)

-  Readme
-  View license
-  Security policy
-  251 stars
-  18 watching
-  57 forks

Releases2

 R2022a Latest
on Mar 15

+ 1 release

Packages

No packages published

Languages

 MATLAB 100.0%

MATLAB Deep Learning Model Hub

Discover pretrained models for deep learning in MATLAB.

Models

Computer Vision

- Image Classification
- Object Detection
- Semantic Segmentation
- Instance Segmentation
- Image Translation
- Pose Estimation
- Video Classification
- Text Detection & Recognition

Natural Language Processing

- Transformers

Audio

- Audio Embeddings
- Sound Classification
- Pitch Estimation
- Speech to Text

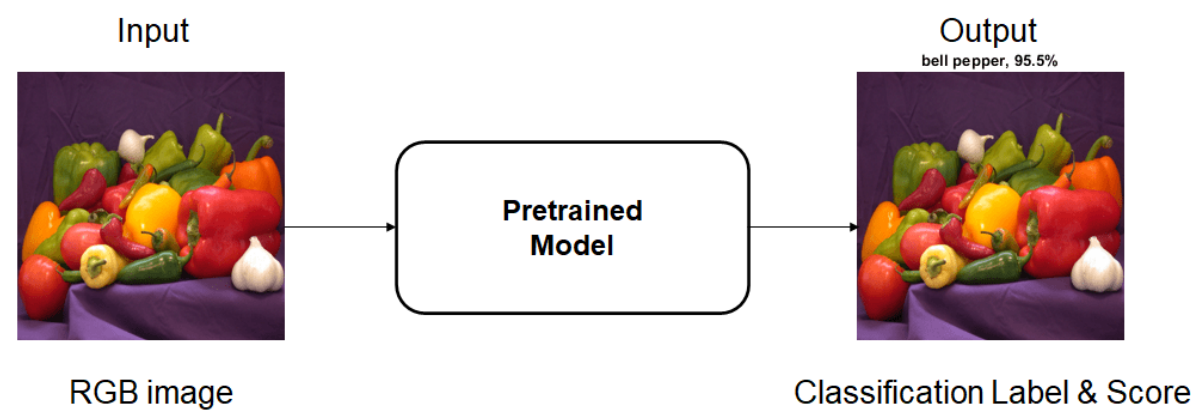
Lidar

- Point Cloud Classification
- Point Cloud Segmentation
- Point Cloud Object Detection

Image Classification

Pretrained image classification networks have already learned to extract powerful and informative features from natural images. Use them as a starting point to learn a new task using transfer learning.

Inputs are RGB images, the output is the predicted label and score:



These networks have been trained on more than a million images and can classify images into 1000 object categories.

Models available in MATLAB:

Network	Size (MB)	Classes	Accuracy %	Location
googlenet	27	1000	66.25	Doc GitHub
squeezenet	5.2	1000	55.16	Doc
alexnet	227	1000	54.10	Doc
resnet18	44	1000	69.49	Doc GitHub
resnet50	96	1000	74.46	Doc GitHub
resnet101	167	1000	75.96	Doc GitHub
mobilenetv2	13	1000	70.44	Doc GitHub
vgg16	515	1000	70.29	Doc
vgg19	535	1000	70.42	Doc
inceptionv3	89	1000	77.07	Doc
inceptionresnetv2	209	1000	79.62	Doc
xception	85	1000	78.20	Doc
darknet19	78	1000	74.00	Doc
darknet53	155	1000	76.46	Doc
densenet201	77	1000	75.85	Doc
shufflenet	5.4	1000	63.73	Doc
nasnetmobile	20	1000	73.41	Doc
nasnetlarge	332	1000	81.83	Doc
efficientnetb0	20	1000	74.72	Doc
ConvMixer	7.7	10	-	GitHub

Tips for selecting a model

Pretrained networks have different characteristics that matter when choosing a network to apply to your problem. The most important characteristics are network accuracy, speed, and size. Choosing a network is generally a tradeoff between these characteristics. The following figure highlights these tradeoffs:

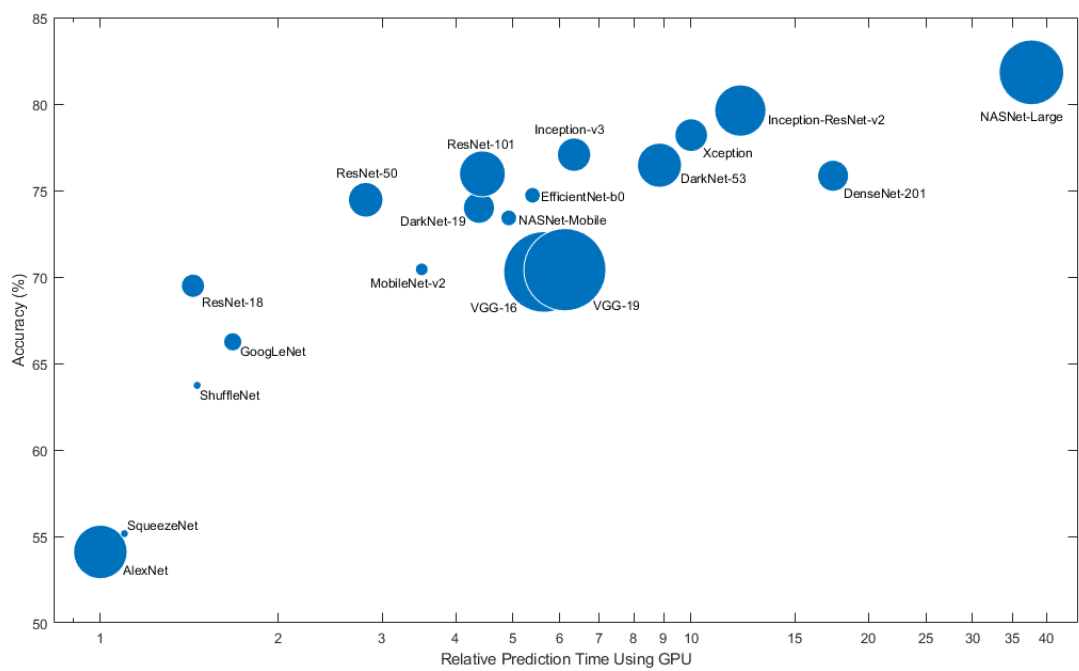


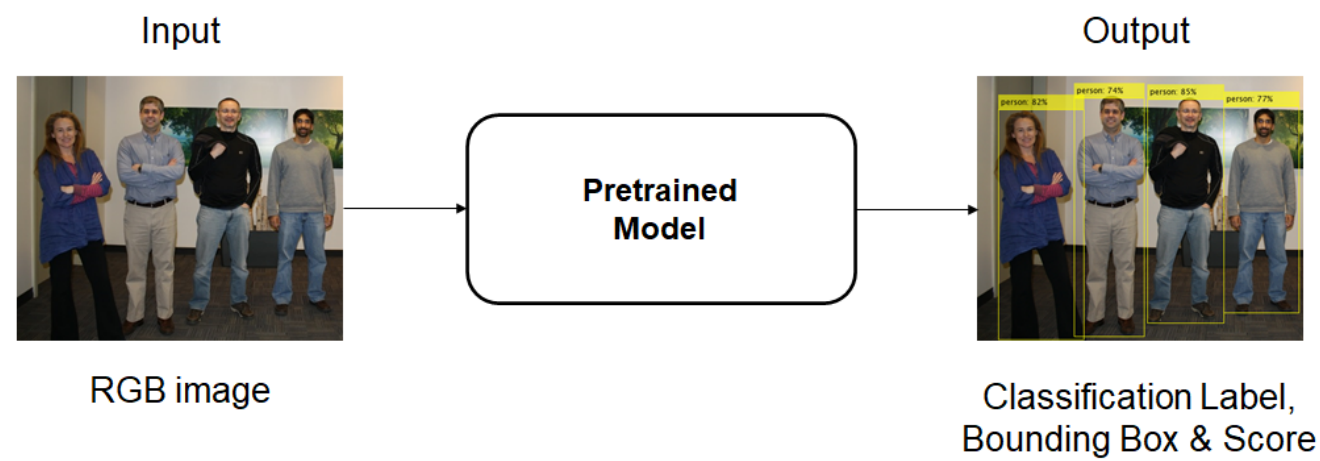
Figure. Comparing image classification model accuracy, speed and size.

[Back to top](#)

Object Detection

Object detection is a computer vision technique used for locating instances of objects in images or videos. When humans look at images or video, we can recognize and locate objects of interest within a matter of moments. The goal of object detection is to replicate this intelligence using a computer.

Inputs are RGB images, the output is the predicted label, bounding box and score:



These networks have been trained to detect 80 objects classes from the COCO dataset. These models are suitable for training a custom object detector using transfer learning.

Network	Backbone Networks	Size (MB)	Mean Average Precision (mAP)	Object Classes	Location
EfficientDet-D0	efficientnet	15.9	33.7	80	GitHub
YOLOX	YoloX-s	32	39.8	80	GitHub
	YoloX-m	90.2	45.9		
	YoloX-l	192.9	48.6		
YOLO v4	yolov4-coco	229	44.2	80	Doc GitHub
	yolov4-tiny-coco	21.5	19.7		
YOLO v3	darknet53-coco	220.4	34.4	80	Doc
	tiny-yolov3-coco	31.5	9.3		

Network	Backbone Networks	Size (MB)	Mean Average Precision (mAP)	Object Classes	Location
YOLO v2	darknet19-COCO tiny-yolo_v2-coco	181 40	28.7 10.5	80	GitHub

Tips for selecting a model

Pretrained object detectors have different characteristics that matter when choosing a network to apply to your problem. The most important characteristics are mean average precision (mAP), speed, and size. Choosing a network is generally a tradeoff between these characteristics.

Application Specific Object Detectors

These networks have been trained to detect specific objects for a given application.

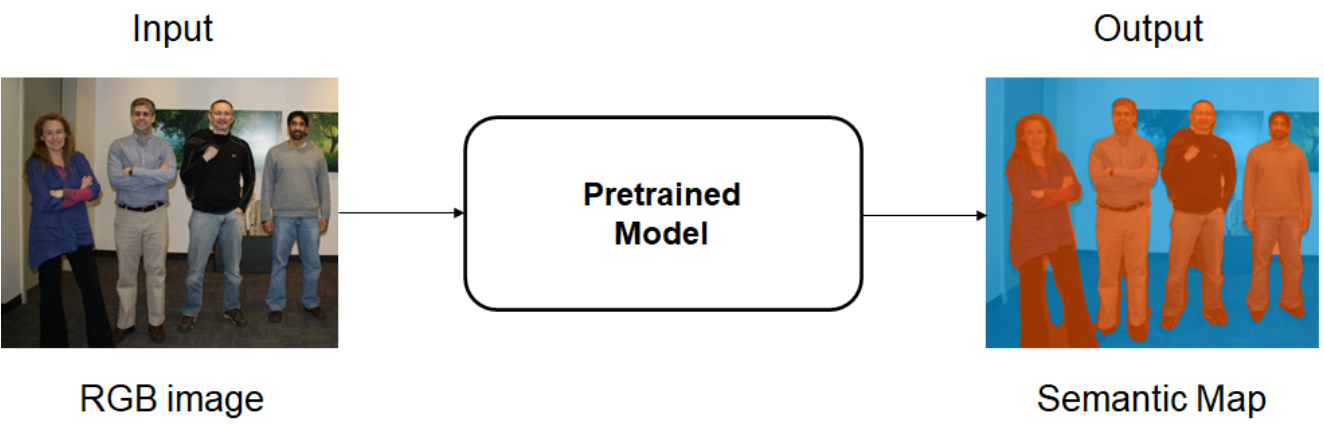
Network	Application	Size (MB)	Location	Example Output
Spatial-CNN	Lane detection	74	GitHub	
RESA	Road Boundary detection	95	GitHub	
Single Shot Detector (SSD)	Vehicle detection	44	Doc	
Faster R-CNN	Vehicle detection	118	Doc	

[Back to top](#)

Semantic Segmentation

Segmentation is essential for image analysis tasks. Semantic segmentation describes the process of associating each pixel of an image with a class label, (such as flower, person, road, sky, ocean, or car).

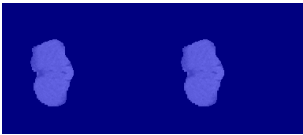
Inputs are RGB images, outputs are pixel classifications (semantic maps).



This network has been trained to detect 20 objects classes from the PASCAL VOC dataset:

Network	Size (MB)	Mean Accuracy	Object Classes	Location
DeepLabv3+	209	0.87	20	GitHub

Application Specific Semantic Segmentation Models

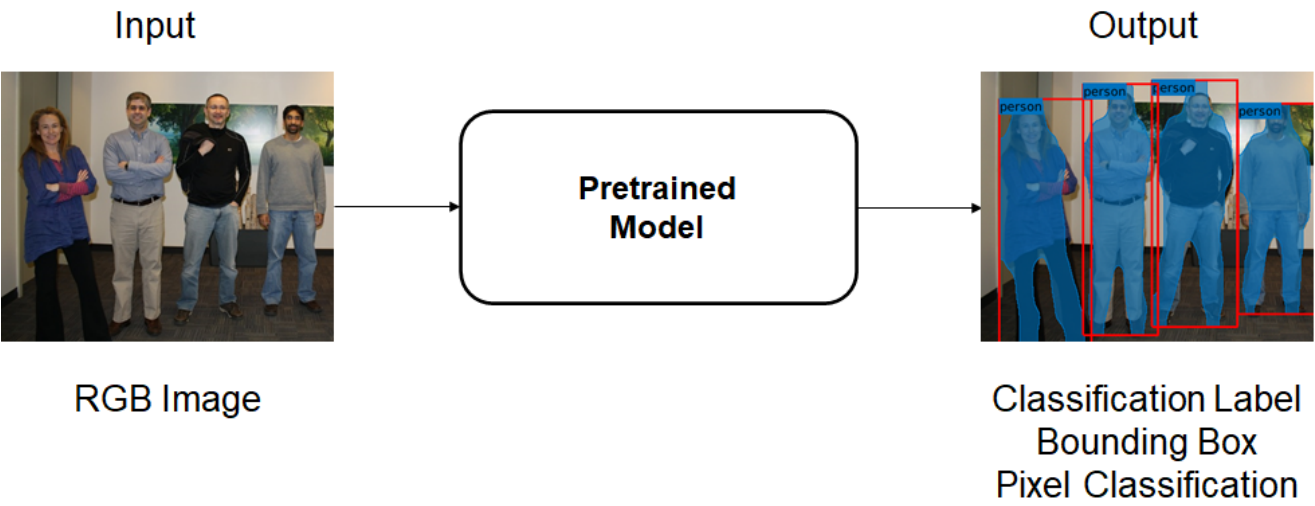
Network	Application	Size (MB)	Location	Example Output
U-net	Raw Camera Processing	31	Doc	
3-D U-net	Brain Tumor Segmentation	56.2	Doc	
AdaptSeg (GAN)	Model tuning using 3-D simulation data	54.4	Doc	

[Back to top](#)

Instance Segmentation

Instance segmentation is an enhanced type of object detection that generates a segmentation map for each detected instance of an object. Instance segmentation treats individual objects as distinct entities, regardless of the class of the objects. In contrast, semantic segmentation considers all objects of the same class as belonging to a single entity.

Inputs are RGB images, outputs are pixel classifications (semantic maps), bounding boxes and classification labels.



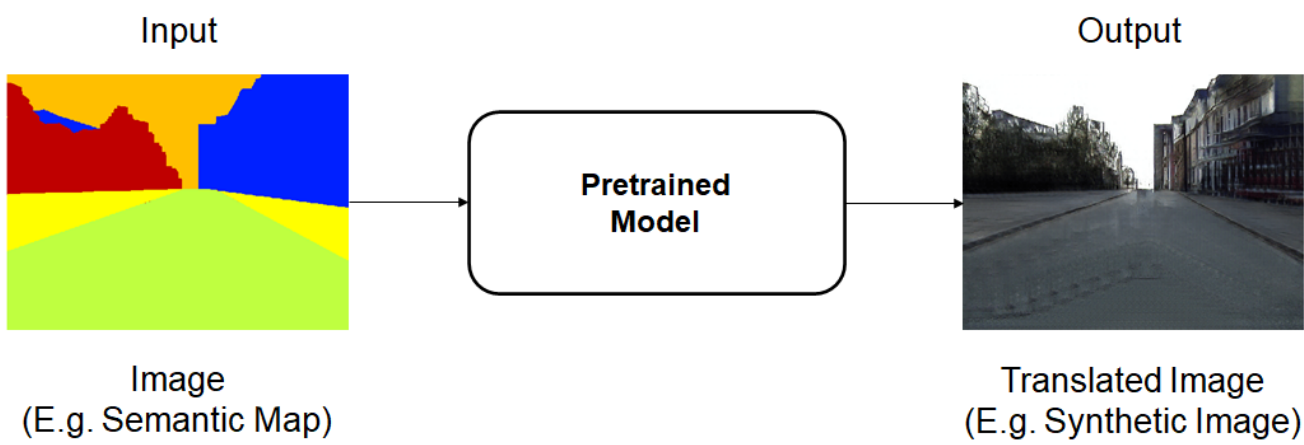
Network	Object Classes	Location
Mask R-CNN	80	Doc Github

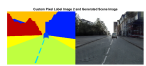
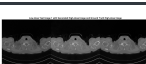
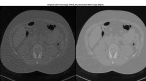
[Back to top](#)

Image Translation

Image translation is the task of transferring styles and characteristics from one image domain to another. This technique can be extended to other image-to-image learning operations, such as image enhancement, image colorization, defect generation, and medical image analysis.

Inputs are images, outputs are translated RGB images. This example workflow shows how a semantic segmentation map input translates to a synthetic image via a pretrained model (Pix2PixHD):



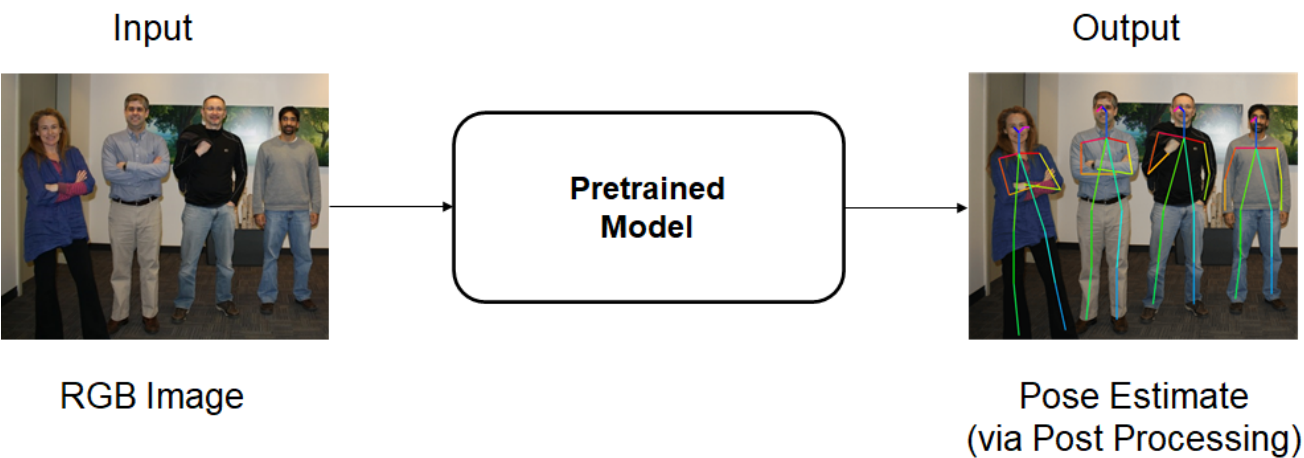
Network	Application	Size (MB)	Location	Example Output
Pix2PixHD(CGAN)	Synthetic Image Translation	648	Doc	
UNIT (GAN)	Day-to-Dusk Dusk-to-Day Image Translation	72.5	Doc	
UNIT (GAN)	Medical Image Denoising	72.4	Doc	
CycleGAN	Medical Image Denoising	75.3	Doc	
VDSR	Super Resolution (estimate a high-resolution image from a low-resolution image)	2.4	Doc	

[Back to top](#)

Pose Estimation

Pose estimation is a computer vision technique for localizing the position and orientation of an object using a fixed set of keypoints.

All inputs are RGB images, outputs are heatmaps and part affinity fields (PAFs) which via post processing perform pose estimation.



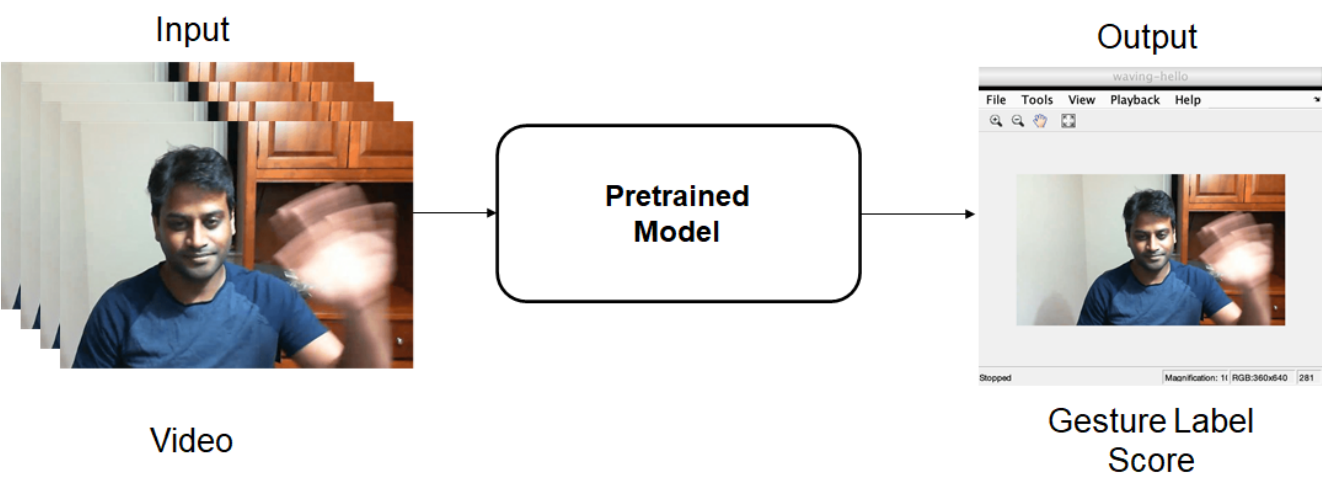
Network	Size (MB)	Location
OpenPose	14	Doc

[Back to top](#)

Video Classification

Video classification is a computer vision technique for classifying the action or content in a sequence of video frames.

All inputs are Videos only or Video with Optical Flow data, outputs are gesture classifications and scores.



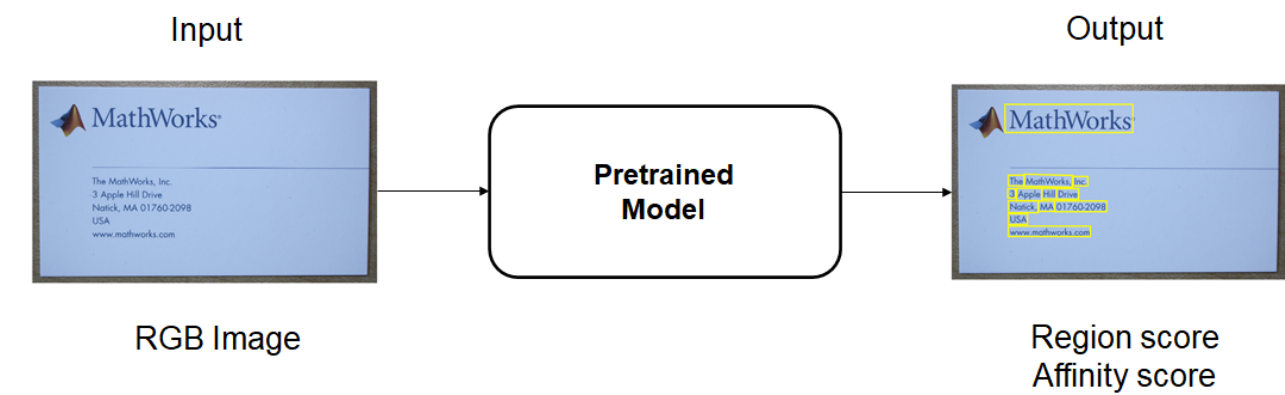
Network	Inputs	Size(MB)	Classifications (Human Actions)	Description	Location
SlowFast	Video	124	400	Faster convergence than Inflated-3D	Doc
R(2+1)D	Video	112	400	Faster convergence than Inflated-3D	Doc
Inflated-3D	Video & Optical Flow data	91	400	Accuracy of the classifier improves when combining optical flow and RGB data.	Doc

[Back to top](#)

Text Detection and Recognition

Text detection is a computer vision technique used for locating instances of text within in images.

Inputs are RGB images, outputs are bounding boxes that identify regions of text.



Network	Application	Size (MB)	Location
CRAFT	Trained to detect English, Korean, Italian, French, Arabic, German and Bangla (Indian).	3.8	Doc GitHub

Application Specific Text Detectors

Network	Application	Size (MB)	Location	Example Output
Seven Segment Digit Recognition	Seven segment digit recognition using deep learning and OCR. This is helpful in industrial automation applications where digital	3.8	GitHub	 7-segment Text Recognition

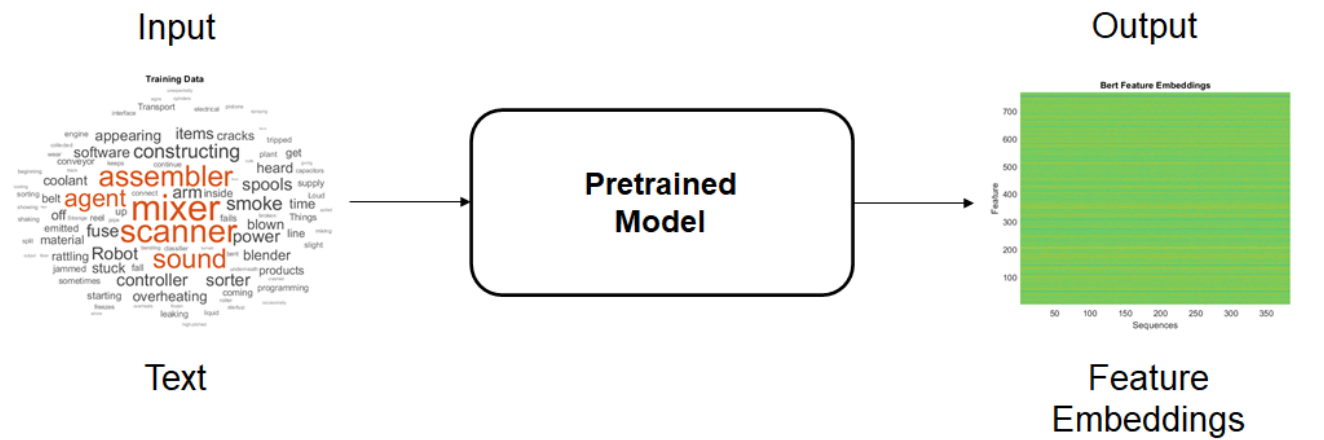
Network	Application	Size (MB)	Location	Example Output
	displays are often surrounded with complex background.			

[Back to top](#)

Transformers (Text)

Transformer pretrained models have already learned to extract powerful and informative features features from text. Use them as a starting point to learn a new task using transfer learning.

Inputs are sequences of text, outputs are text feature embeddings.



Network	Applications	Size (MB)	Location
BERT	Feature Extraction (Sentence and Word embedding), Text Classification, Token Classification, Masked Language Modeling, Question Answering	390	GitHub

Application Specific Transformers

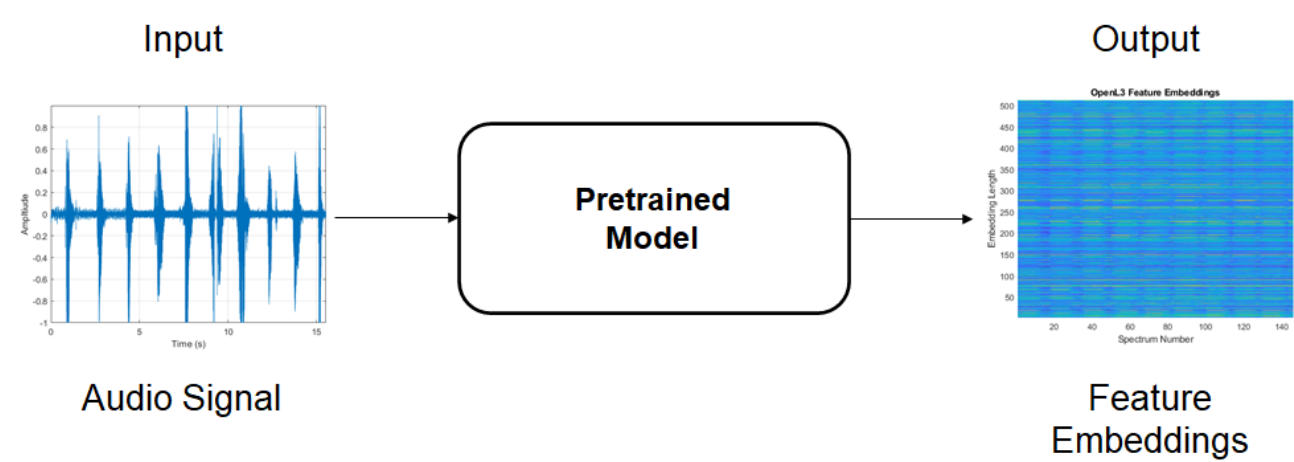
Network	Application	Size (MB)	Location	Output Example
FinBERT	The FinBERT model is a BERT model for financial sentiment analysis	388	GitHub	<pre>>> mdl = finbert; >> str = "Experts estimate the value of its remaining stake in the company at \$ 27 million."; >> tokens = encode(tokenizer,str); >> X = padsequences(tokens,2,"PaddingValue",mdl.Tokenizer.PaddingCode); >> sentiment = finbert.sentimentModel(X,mdl.Parameters) sentiment = categorical neutral</pre>
GPT-2	The GPT-2 model is a decoder model used for text summarization.	1.2GB	GitHub	<pre>>> inputText = help('magic') >> summary = generateSummary(mdl,inputText) summary = ': Magic square.'</pre>

[Back to top](#)

Audio

Audio Embedding pretrained models have already learned to extract powerful and informative features from audio signals. Use them as a starting point to learn a new task using transfer learning.

Inputs are Audio signals, outputs are audio feature embeddings.



Network	Application	Size (MB)	Location
VGGish	Feature Embeddings	257	Doc
OpenL3	Feature Embeddings	200	Doc

Application Specific Audio Models

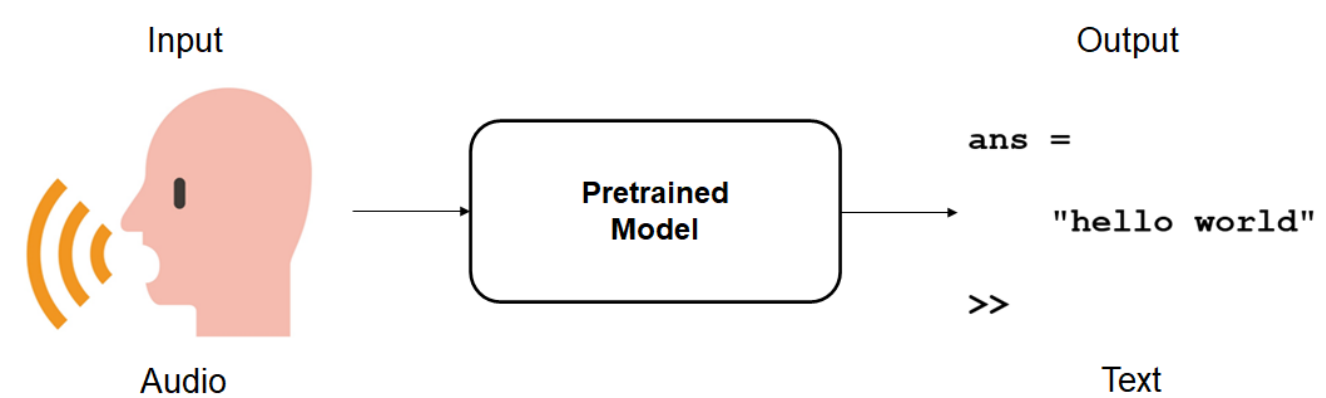
Network	Application	Size (MB)	Output Classes	Location	Output Example
YAMNet	Sound Classification	13.5	521	Doc	
CREPE	Pitch Estimation (Regression)	132	-	Doc	

☰ [README.md](#)

Speech to Text

Speech to text pretrained models take an audio input and translate it into a text output. They are useful in digitizing audio files for downstream text processing tasks such as text summarization and sentiment analysis.

Inputs are Audio signals, outputs is text.



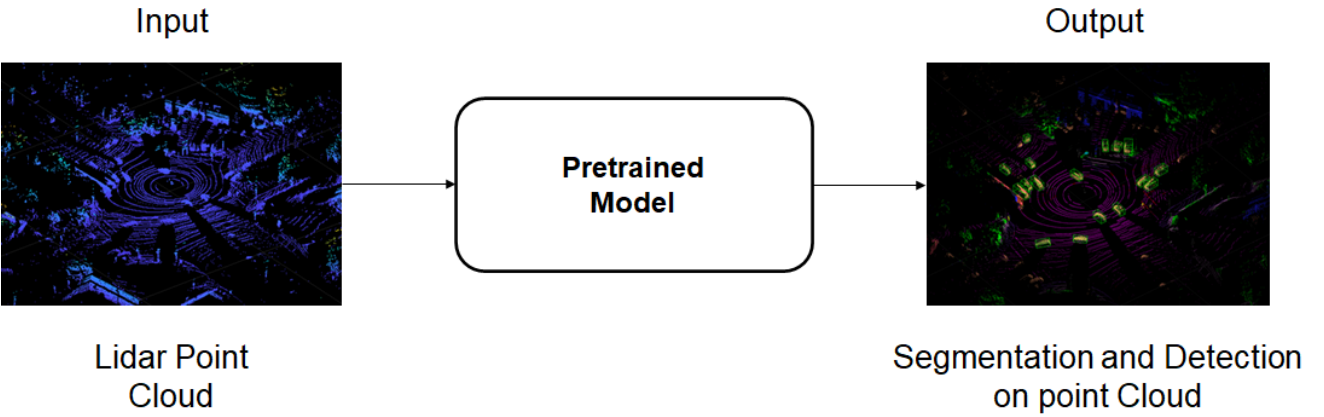
Network	Application	Size (MB)	Word Error Rate (WER)	Location
wav2vec	Speech to Text	236	3.2	GitHub
deepspeech	Speech to Text	167	5.97	GitHub

[Back to top](#)

Lidar

Point cloud data is acquired by a variety of sensors, such as lidar, radar, and depth cameras. Training robust classifiers with point cloud data is challenging because of the sparsity of data per object, object occlusions, and sensor noise. Deep learning techniques have been shown to address many of these challenges by learning robust feature representations directly from point cloud data.

Inputs are Lidar Point Clouds converted to five-channels, outputs are segmentation, classification or object detection results overlayed on point clouds.



Network	Application	Size (MB)	Object Classes	Location
PointNet	Classification	5	14	Doc
PointNet++	Segmentation	3	8	Doc
PointSeg	Segmentation	14	3	Doc
SqueezeSegV2	Segmentation	5	12	Doc
SalsaNext	Segmentation	20.9	13	GitHub
PointPillars	Object Detection	8	3	Doc
Complex YOLO v4	Object Detection	233 (complex-yolov4) 21 (tiny-complex-yolov4)	3	GitHub

[Back to top](#)

Model requests

If you'd like to request MATLAB support for additional pretrained models, [please create an issue from this repo](#).

Alternatively send the request through to:

[David Willingham](#)
Deep Learning Product Manager
dwilling@mathworks.com

Copyright 2022, The MathWorks, Inc.